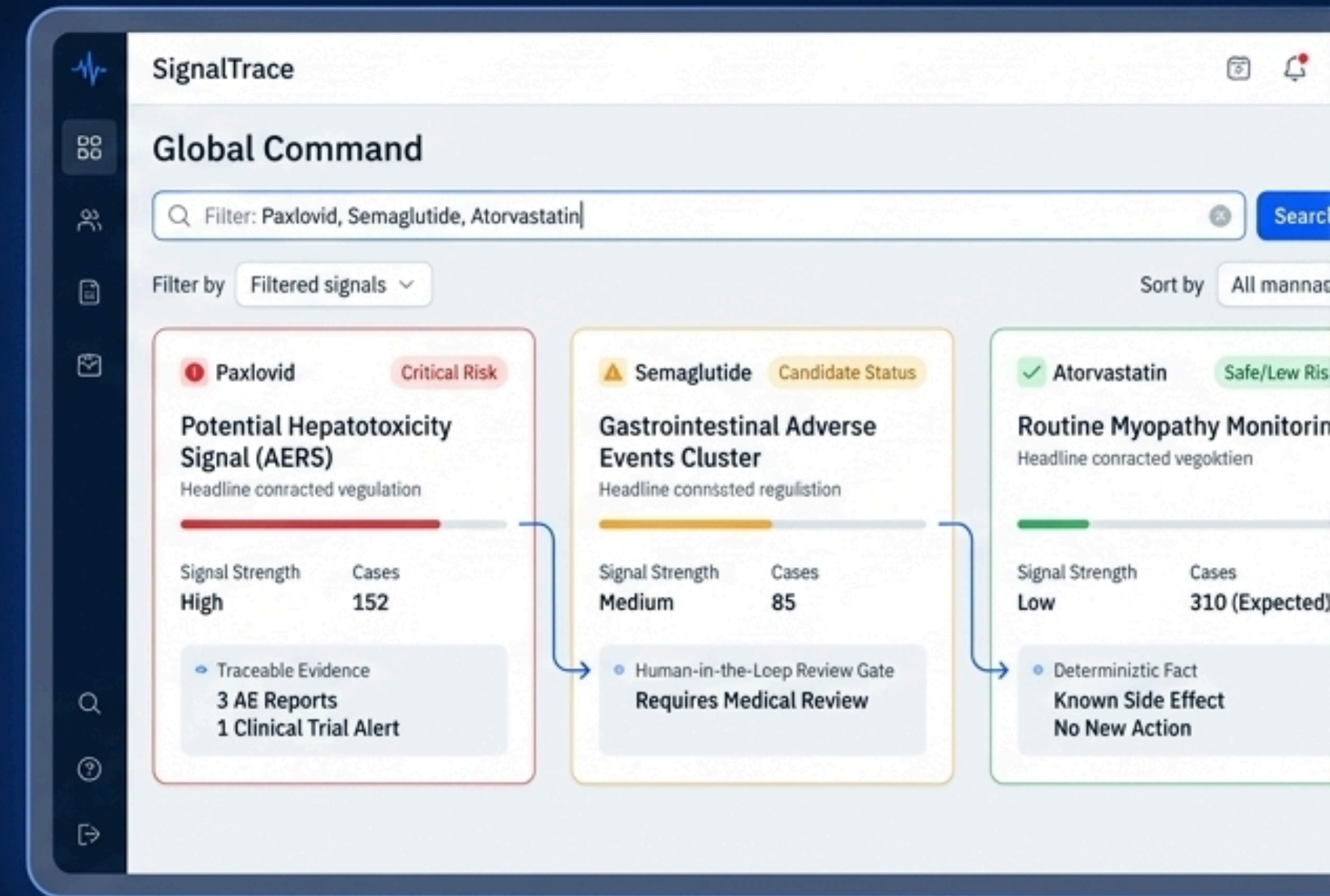


SignalTrace

From Safety Signal to Review-Ready Evidence

AI-assisted pharmacovigilance decision support for turning drug-safety signals into structured, explainable and traceable evidence.



Finding a Signal Is Only the Beginning

Pharmacovigilance teams detect candidate drug-event signals from vast datasets.

Source

20M+
FAERS adverse-event reports

Detection

Candidate Signal

The Evidence Thread

Teams manually validate case quality, check duplicates, interpret statistics, and trace signals to documents.

Manual Investigation

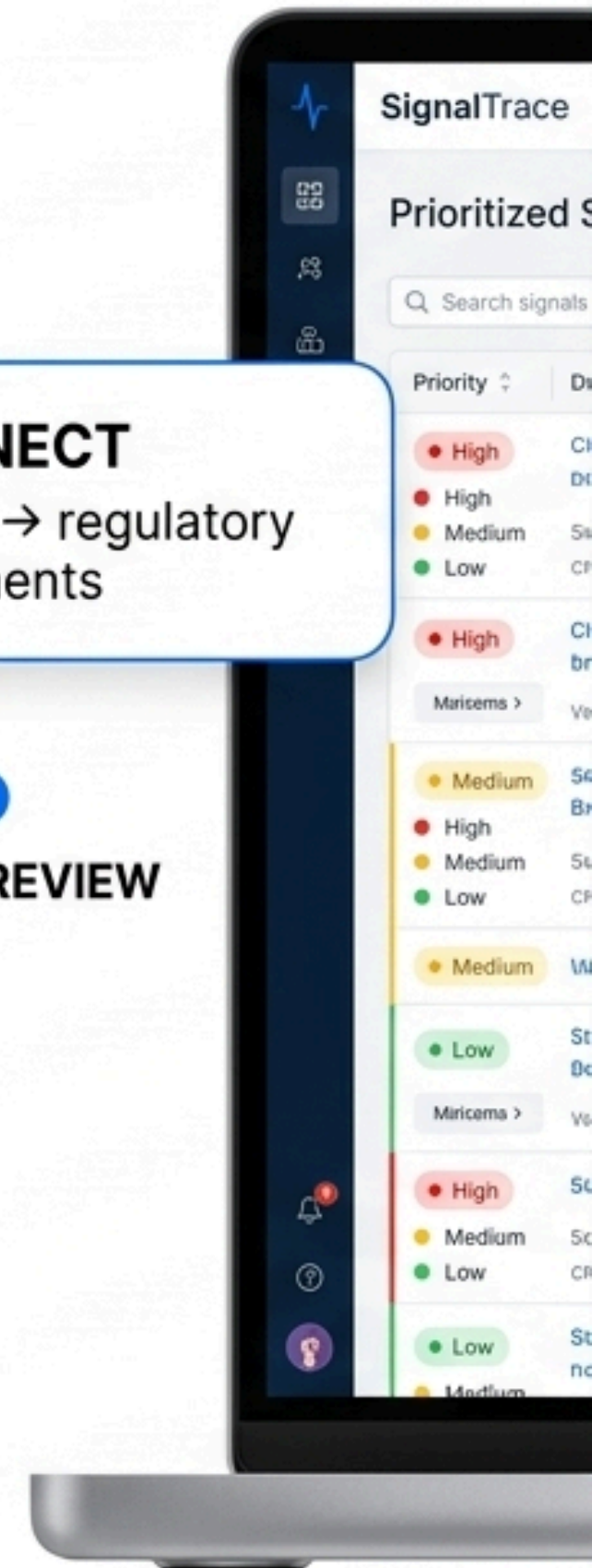
Burdened drug safety reviewers and regulatory affairs teams.

Destination

Regulatory Review

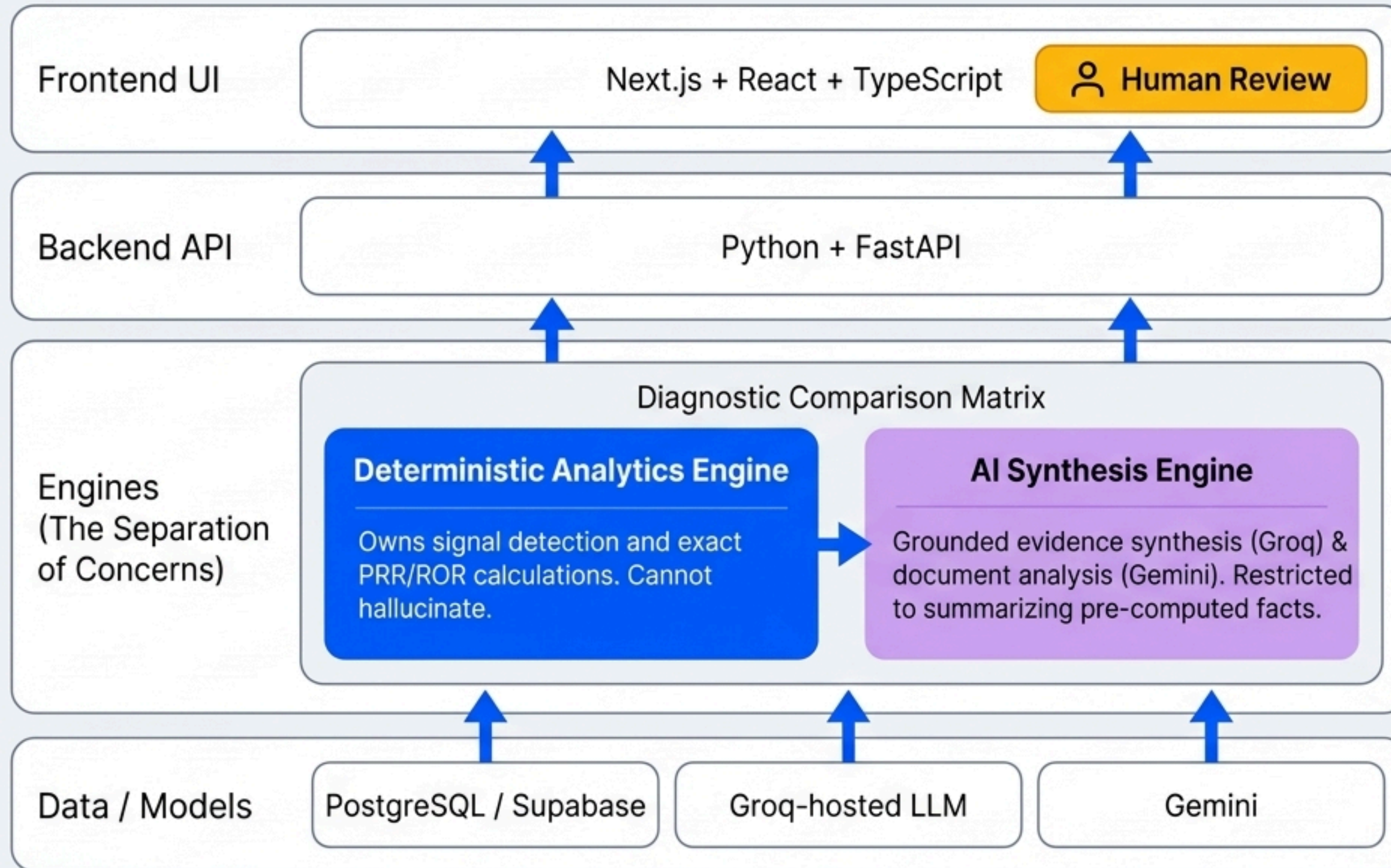
**The bottleneck is not finding data.
It is turning data into review-ready evidence.**

An AI-Assisted Pharmacovigilance Decision-Support Platform



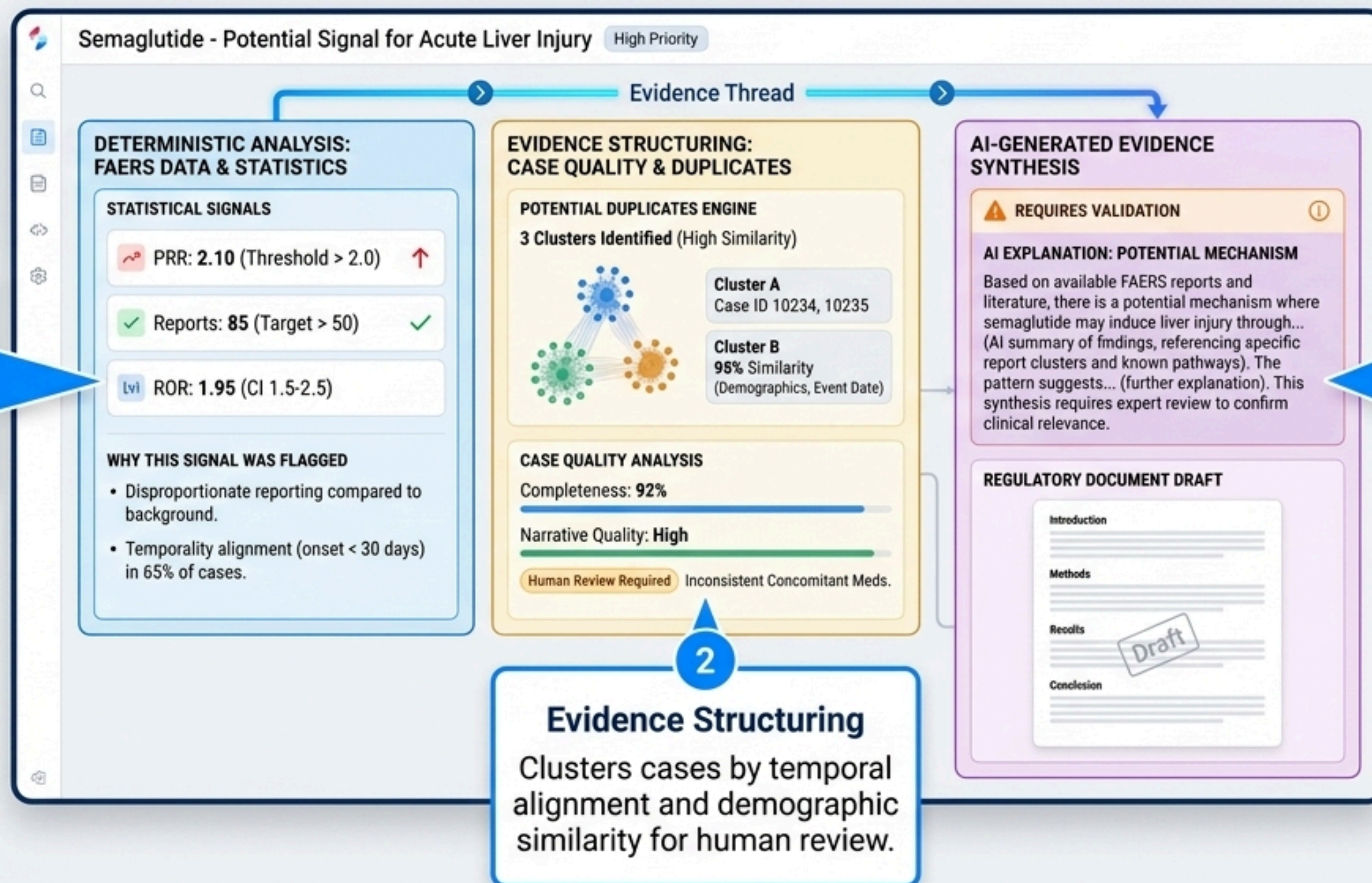
■ Calculated / Deterministic
 ■ AI-Assisted Synthesis
 ■ Human Review Gate

Architecture Built for Traceability

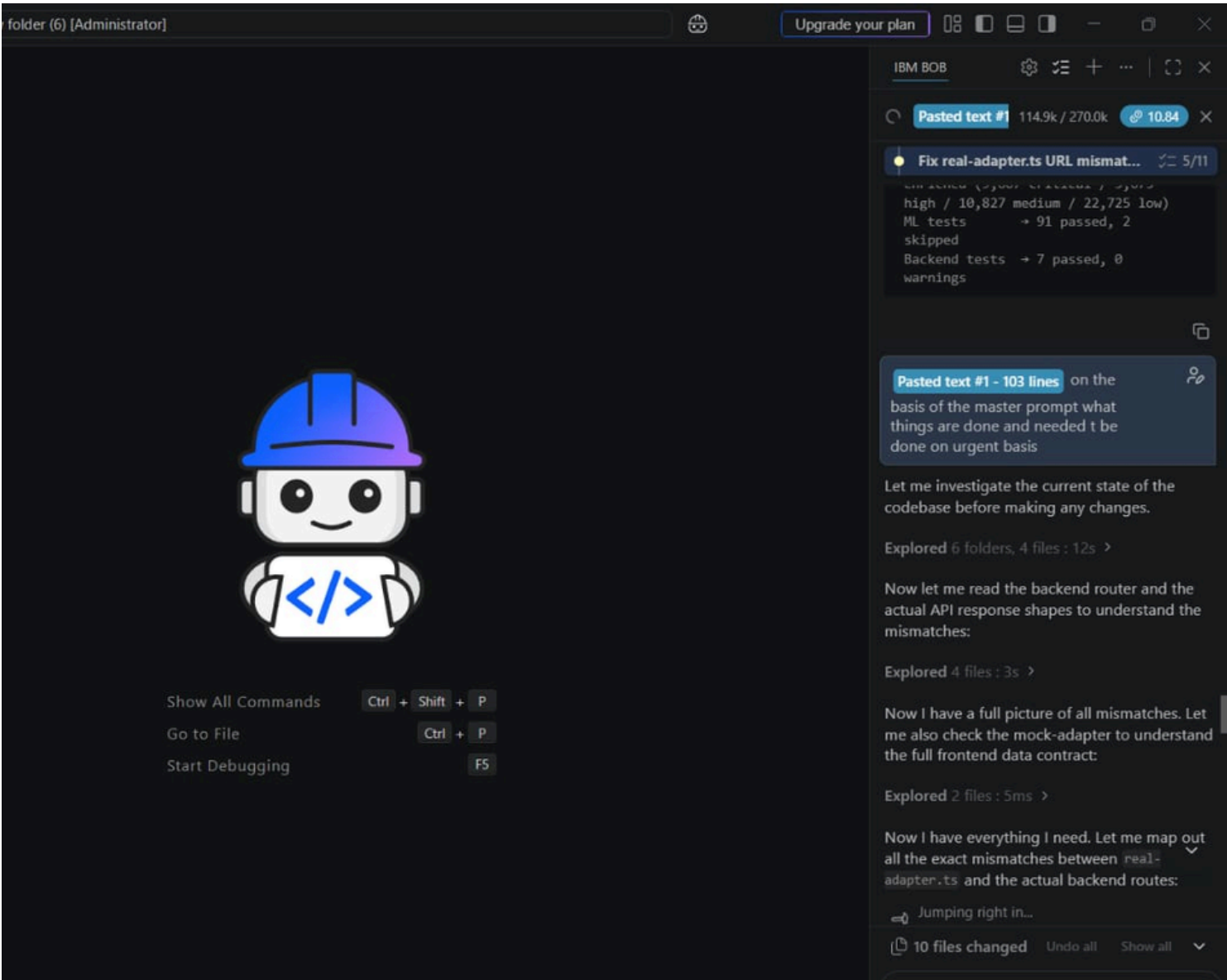
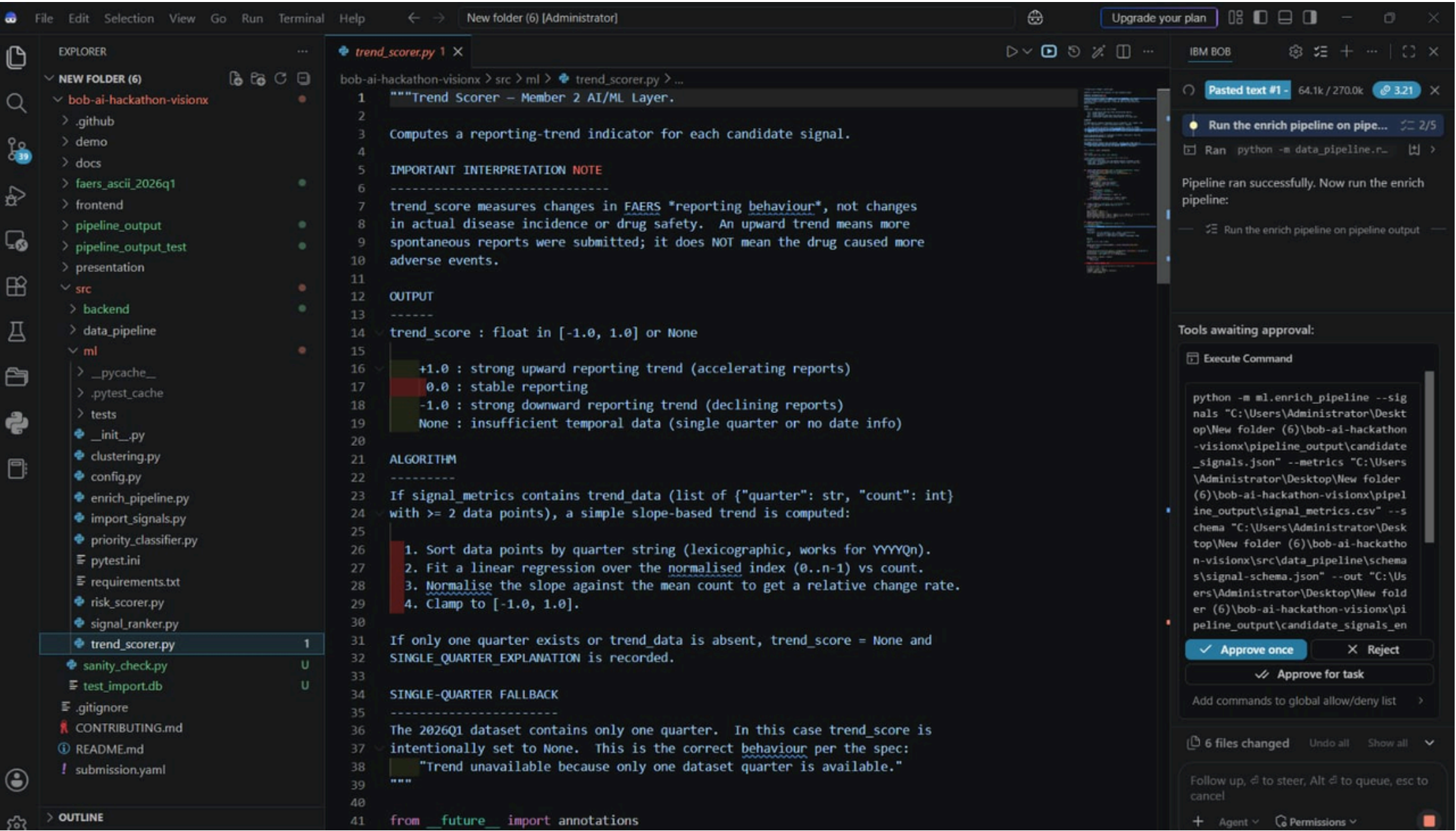


- Separation of Concerns: Analytics engines generate statistics; LLMs synthesize those specific facts.
- **Grounding:** AI cannot invent statistics; it reads strictly from the deterministic database.
- Traceability: Every AI output requires a human-in-the-loop decision gate.

From Queue to Review-Ready Evidence



Usage of IBM Bob



IBM Bob: The Engineering Accelerator



IBM Bob acting as an AI coding assistant during frontend engineering.

Accelerated Refactoring

Component Isolation

Refactored large UI blocks into reusable React components.

Network Resilience

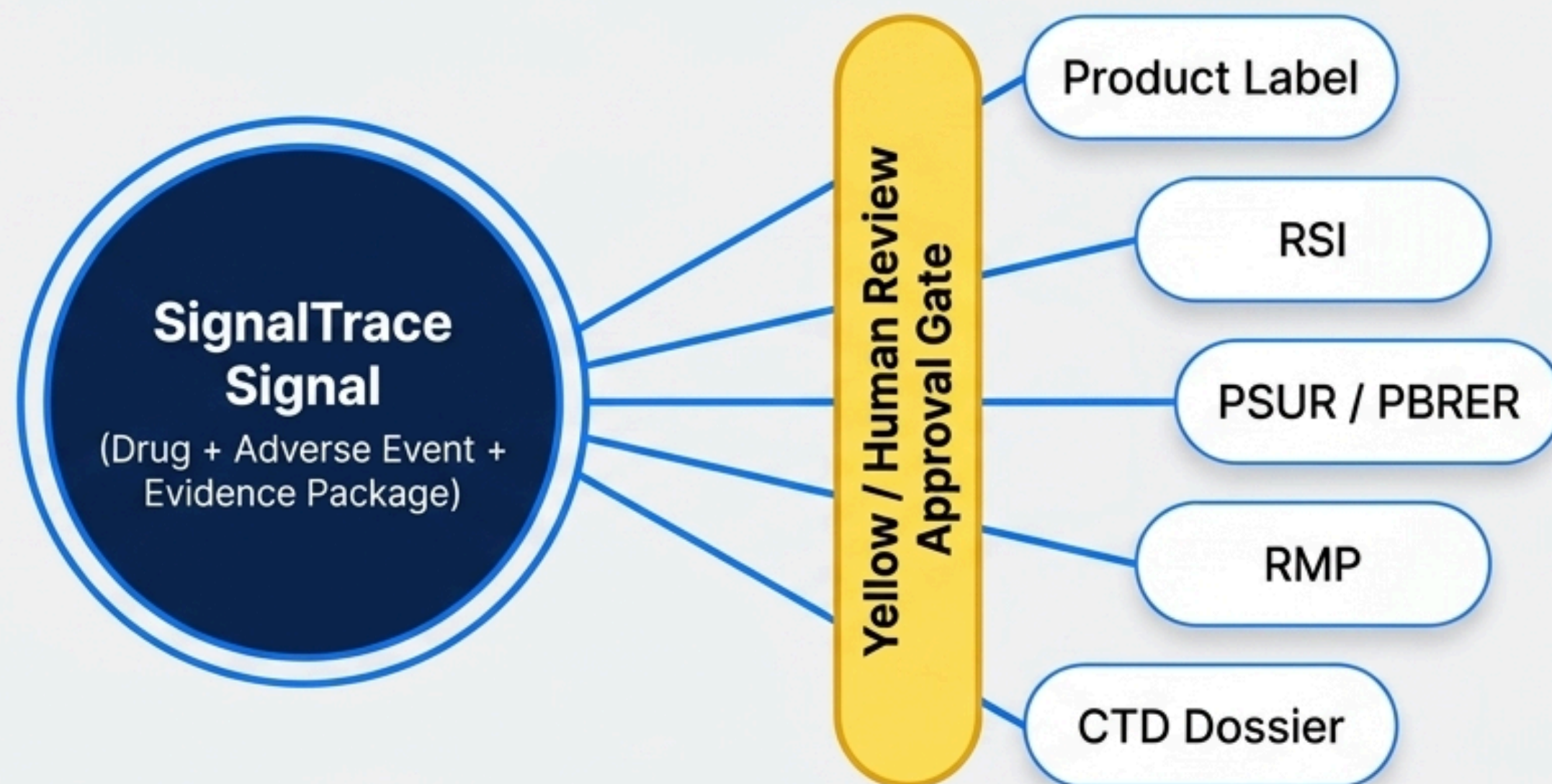
Isolated API logic behind a unified ApiClient pattern.

Demo Hardening

Built type-safe MockAdapters for predictable demo data, handling loading, empty, and error states reliably.

Outcome: Faster iteration, a cleaner architecture, and a highly resilient, enterprise-grade demo experience.

The Signal-to-Regulatory Bridge



Faster Review: Eliminates repetitive, manual evidence-compilation work.

Traceability: Preserves a clear, auditable path from raw signal to regulatory document.

Scalability: Architecture ready to expand beyond FAERS to literature and clinical trials.

Human Expertise: Experts spend time on clinical judgment, not data entry.

Turning fragmented safety data into a review-ready evidence trail.

The Builders

HARSHIL

THAKKAR

Data Engineering
& Feature
Pipeline

SHARNAM

KATWALA

AI / ML & Signal
Intelligence

MANN SHAH

Backend & NLP /
Regulatory
Intelligence

JIYA SADARIA

Frontend,
UI/UX

Project: SignalTrace

Track: IBM Bob AI Hackathon / Pharmacovigilance & Regulatory Intelligence

Team: VisionX

“SignalTrace turns a candidate safety signal into a traceable, review-ready evidence trail — while keeping humans in control.”